Memorandum

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To: Dr. Reese and Dr. Jeong

From: Joshua Dixon, Danielle Edington, Miguel Martinez-Garcia, and Diana Mejia

CC: Charles Gould, Dr. Srivastava, and Aluel Go

Re: Assignment 6 – Design Alternatives and Decision Matrix

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This report outlines an overview of ongoing work to develop an optimization model for agrivoltaic systems in Michigan's Districts 7, 8, and 13 as illustrated in Figure F1 and Table F1 of Appendix F (Michigan State University Extension, n.d.). The model is designed to serve as an economic decision-making tool that balances crop yields with photovoltaic energy generation, using predefined infrastructure, environmental, financial, time, and crop parameters. By integrating farmer needs with technical constraints, the tool aims to increase income through optimal land allocation between solar arrays and crops. Future efforts will refine the model by defining representative farm profiles, evaluating array configurations, validating results with real-world data, and identifying applicable state and federal incentives to further enhance profitability and long-term dual-use application.

BE 485 Biosystems Design Techniques

Assignment #6 Design Alternatives and Decision Matrix

Optimizing Agrivoltaic Systems for Crop Yield and Energy Production in Michigan

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List of Acronyms, Abbreviations, Initialisms, Units

Acronyms, Abbreviations, & Initialisms:

FSMA	food safety modernization act
IEEE	institute of electrical and electronics engineers
ISO	international organization for standardization
LCOE	levelized cost of electricity
LER	land equivalence ratio
LL	land loss
MIOSHA	Michigan occupational safety and health administration
NEC	national electrical code
NPV	net present value

PAR	photosynthetically active radiation
PV	photovoltaics
PVWatts	photovoltaic watts
SAMS	system advisor model
SOM	soil organic matter
Units:	
°	degree of panel orientation
°C	temperature, degrees Celsius
°F	temperature, degrees Fahrenheit
ft	feet
ft^2	square feet
g CO_2e/kWh	grams carbon dioxide emission per kilowatt-hours
mi	miles

Introduction and Project Establishment

Recent shifts in global energy supply systems have introduced photovoltaics as a new competitor for land traditionally dedicated to agricultural production, prompting important discussions about dual land-use strategies (Nonhebel, 2005). Michigan State University Extension, represented by Charles Gould, has been at the forefront of these discussions exploring how utility-scale solar developments can be integrated with agriculture. This project focuses on Michigan's agricultural regions. Specifically districts 7, 8, and 13 are shown in Figure A1 of Appendix A, where farmland is expected to increasingly intersect with renewable energy infrastructure. These areas are frequently targeted for solar development because they tend to be relatively flat, clear, and well-drained with access to nearby electrical infrastructure (Kinzer et al., 2025). Conversations with farmers have revealed that many had not considered the possibility of growing crops or grazing livestock within solar projects (Michigan State University College of Agriculture & Natural Resources, 2024). This highlights that while solar development is expanding across Michigan, the practice of combining crop production with photovoltaic energy generation remains overlooked.

Successful agrivoltaic project requires early coordination between farmers and solar developers to mitigate soil compaction and topsoil loss during installation, ensure proper spacing and height of arrays, maintain adequate headlands, and design cropping systems that align with equipment access and management needs (Michigan State University College of Agriculture & Natural Resources, 2024). This offers opportunities for farmers to manage vegetation through cropping or grazing while generating additional income streams. Farmers are already knowledgeable in vegetation management, and their involvement in solar projects can transform what is commonly outsourced to landscaping companies (Kinzer et al., 2025). From this, developers receive reliable

vegetation management, farmers gain supplemental income, and communities benefit from both electricity production and the preservation of land.

Evaluating these systems often relies on metrics such as the Land Equivalence Ratio (LER), which measures combined energy and crop productivity demonstrated in Figure 1 (Warmann, et al., 2024). It provides a starting point for assessing agrivoltaic performance. LER is a nondimensional metric that is equal to the amount of land required to reproduce the agrivoltaic system's energy and crop yield using conventional solar energy and agricultural practices on separate land. The reduction of land used for agricultural production is reduced by a land loss factor (LL), which is a value between 0 and 1, representing the percentage of land lost to agrivoltaic structure and buffer zones.

$$LER = \frac{Yield_{Crop,AV}}{Yield_{Crop,conv}} \cdot (1 - LL) + \frac{Yield_{elec,AV}}{Yield_{elec,conv}} \qquad (1)$$

This project then seeks to evaluate agrivoltaic system design and implementation in Michigan, guided by both farmer perspectives and established research on dual-use solar practices.

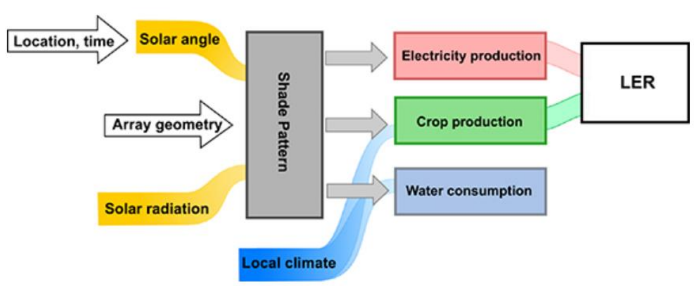

Figure 1: Schematic representation of agrivoltaics system to determine land equivalence ratio (LER) (Warmann et al., 2024)

Justification

Life-cycle assessments confirm that solar photovoltaic (PV) systems form a low-carbon alternative to fossil fuel electricity (National Renewable Energy Laboratory, 2021; Pacca et al.,

2007). A synthesis of electricity-generation technologies reports median life-cycle greenhouse gas emissions of about 57 g CO_2e/kWh for solar PV and much higher values for natural gas and coal (National Renewable Energy Laboratory, 2021). These results support ongoing expansion of PV as part of regional decarbonization and show that agrivoltaic configurations can place this low-carbon generation directly on cash-crop land.

Research on solar-park development shows that construction of traffic and temporary access roads compact soil, reduce infiltration, and impair long-term soil quality on agricultural land (Welsh Government & ADAS, 2023). Ecological field work shows that partial shading by solar panels can delay blooming, increase late-season floral abundance for pollinators, and produce only modest declines in grassland productivity when communities adjust to lower light (Graham et al., 2021; Kannenberg et al., 2023). Economic studies and benchmarking report capital-cost premiums for agrivoltaic systems, limited incremental revenue, and profitability that is sensitive to hardware costs and energy prices (National Renewable Energy Laboratory, 2020; Strub et al., 2024). Studies of social acceptance of dual land use report more favorable views when projects maintain visible agricultural production (Pascaris et al., 2022; Nesbit et al., 2023). Grid and policy analyses in the Great Lakes region describe zoning ordinances, setbacks, and permitting rules that limit feasible PV capacity and raise costs (Owusu-Obeng et al., 2022). Comparing agrivoltaic configurations can identify layouts that protect soil function, support pollinators, fit economic, social, and siting constraints, and still provide meaningful PV capacity on cash-crop land.

For farmers growing cash crops in mid-Michigan, there are a lot of choices to make around agrivoltaics. It is necessary to have accessible tools that support farmers in these choices.

Problem Statement

Create an optimization model that determines the optimal balance of land-use between crop productivity and photovoltaic energy generation (agrivoltaics), and guides Michigan specialty crop farmers in maximizing profit.

Objectives

To guide the development of a financially viable agrivoltaics system for Michigan farms. A set of core objectives were identified. These focus on quantifying energy potential, validating economic performance, and ensuring practical integration with existing agricultural operations.

- Develop a clear understanding of vegetable specialty crop farming production systems, geographic locations, and management practices in Michigan to identify and define a representative target farm profile for agrivoltaic system design and evaluation
- Gain a comprehensive understanding of fixed tilt, vertical bifacial, and single axis tracking solar arrays through collaboration with contract manufacturers, solar producers, and simulation tools such as SAMS or PVWatts
- Design a model that aligns with the target farm profile and optimal solar array configuration
- Test the developed model in a real-world application, validating with coefficients and parameters before verifying results from Michigan solar installer data
- Identify and evaluate federal and state incentives or specific funding (USDA, NRCS, etc.) programs for Michigan farmers to implement agrivoltaics

Constraints

The agrivoltaics design is set by constraints that define the feasibility and scope of the project. Constraints have been broken into 5 categories: infrastructure, environmental, financial, time, and crop selection. Each category contributes to the optimal design for Michigan's farmers and solar energy conditions. Infrastructure provides a mechanical overview such as array height, availability of space, and layouts. Environmental considers the weather and agriculture parameters in Michigan regions. Financial reflects the economic analysis to be considered in introducing photovoltaics into farms. Time addresses the projects' time of completion, as well as

post construction site stabilization. Finally, crop directs crop focus based on current and future limitations.

Infrastructure

- Minimum 18 ft between solar array lines, subdivided into three 6 ft crop beds, each with three crop rows (Mamun et al., 2022)
- Arrays must be elevated at least 6-8 ft above ground to allow airflow and tractor access (Mamun et al., 2022)
- Arrays must be connected to grid infrastructure within 0.5 mi to minimize transmission losses and costs
- Lifespan of solar array ranges from 15-25 years
- Tile drainage and water management

Environmental

- Soil pH must remain within 6.0–7.5 for proper crop growth depending on the region
- Systems must withstand Michigan's annual temperature range of 0°F–95°F (-17°C to 35°C)
- Sunlight intensity under modules must not fall below 50% PAR (photosynthetically active radiation) over 1.0 ft^2 area
- At least two cover crop species or pollinator-friendly plants must be integrated per rotation cycle

Financial

- Systematic or legal costs (leases or contracts for the land, permitting and licensing, insurance, and taxes)
- Solar investor partnerships for implementation of solar panels
- Land lease rates must remain competitive with average Michigan cropland rental rates or crop yield

Time

- Estimated prototype project completion: April 2026
- Cover crops or forage must be established within 1 growing season post-construction to stabilize soils

Crop Selection

- Select crops proven to tolerate 30-50% reduced light
- Avoid deep-rooted crops in compacted zones, use shallow rooted vegetables or legumes
- Include legumes at least once every 3 years to restore nitrogen and SOM

Project Background and Deliverables

Agrivoltaic systems present both opportunities and challenges for Michigan farmers. They offer multiple revenue streams through crop yield and PV energy generation. However, their implementation requires careful attention to soil health, equipment compatibility, and long-term farm viability. Construction can often lead to soil compaction, erosion, and organic matter depletion, reducing productivity if not properly managed (Welsh Government & ADAS, 2023).

The optimization model developed in this project is designed to address these technical challenges by aligning with the four soil health principles: minimize disturbance, maximize cover, maximize biodiversity, and maximize living roots (Kinzer et al., 2025). It incorporates siting, design, and operational parameters that protect soil function throughout the solar development life cycle. Web soil survey data and site-specific soil testing will be used to identify areas most susceptible to compaction or erosion. Informing layout recommendations that preserve topsoil, minimize heavy equipment passes, and prioritize vegetation cover. By simulating dual-use scenarios, the model will evaluate how cropping and solar array systems interact. This ensures biodiversity, soil cover, and root life are maintained.

Deliverables will include a linear programming decision-making tool that integrates the identified target farm profile with infrastructure, environmental, crop, and financial constraints. The model will apply the constraints to the decided array configuration specified in the decision matrix. Outputs will provide farmers with clear recommendations on land allocation between crops and solar arrays, maximizing income while protecting soil health and ensuring long-term viability of dual-use sites.

Technical Design Approach

Standard and Regulations

Implementation of agrivoltaic systems on Michigan cash-crop farms must comply with national and state standards, electrical codes, worker-safety regulations, food-safety rules, and land-use laws. Quality management follows ISO 9001 (ISO, 2015a). Environmental management follows ISO 14001 (ISO, 2015b). Occupational health and safety management follows ISO 45001 (ISO, 2018). Electrical design follows IEEE 1547 (IEEE, 2018). Electrical safety in power systems follows the National Electrical Safety Code (IEEE C2) (IEEE, 2022). Photovoltaic installations follow the National Electrical Code (NEC) Article 690 (National Fire Protection Association, 2023). State enforcement uses the 2023 Michigan Electrical Code and the 2021 Michigan Building Code (Michigan Department of Licensing and Regulatory Affairs, Bureau of Construction Codes, 2021, 2023). These frameworks define requirements for grounding, rapid-shutdown capabilities, over-current protection, and structural loading under wind and snow, and shape array layout, racking design, and interconnection options for agrivoltaic systems.

Worker and construction safety follow federal OSHA electric-power and construction standards (Electric power generation, transmission, and distribution, 2014; Safety and health regulations for construction, 2009). Michigan enforces these protections through the Michigan Occupational Safety and Health Act (MIOSHA) (Michigan Occupational Safety and Health Act, 1974). For farms growing food crops beneath and between arrays, the FSMA Produce Safety Rule (21 C.F.R. Part 112) treats panel racking and infrastructure as food-contact equipment and it requires design, cleaning, and maintenance that prevent contamination of produce (Standards for growing, harvesting, packing, and holding of produce for human consumption, 2016). Key Michigan statutes, including the Farmland and Open Space Preservation Act, the Natural

Resources and Environmental Protection Act, the Michigan Zoning Enabling Act, and Public Act 233 of 2023, govern preserved farmland, environmental protection, local zoning authority, and siting of large renewable energy projects (Farmland and Open Space Preservation Act, 1974; Natural Resources and Environmental Protection Act, 1994; Michigan Zoning Enabling Act, 2006; Public Act 233 of 2023, 2023). Together, these standards and regulations define the safety, environmental, and siting constraints used to screen agrivoltaic design alternatives.

Safety Elements

The interaction that is formed between agriculture and photovoltaics creates a combination of operational elements. These elements can make an impact on safety risk assessment in the operational area. The safety elements to consider are environmental safety, electrical structure, and structural and mechanical safety.

Environmental Safety

Agrivoltaic systems impact environmental conditions by increasing precipitation patterns, heavy snow, and high winds. These are risk elements to be considered when implementing the systems. It reinforces the importance of choosing suitable array configuration height, panel design, and tilt angle. Crop spacing is recommended with array columns to ensure farm equipment and growing flow (Lee et al., 2023).

Electrical Safety

Agrivoltaics implementation introduces high voltage equipment into the agricultural field. This implementation can create risks such as electric shock hazard, malfunction and damage of equipment, and fire hazards (Ya'acob et al., 2023). A proper grounding installation of solar panels is essential. It is recommended for farmers to work with solar investors and contractors

that will provide maintenance. This includes regular inspections, identification of failures or deficiencies, and periodic visits to the farms to ensure the quality and safety of the system.

Structural and Mechanical Safety

In this category, risks associated are structural solar panel failures, dust deposition, and accidental farm equipment impact. Safety measures for these risks are proper structural heights, adequate distance between arrays, flexible placement, and system elevation improvements to enhance energy output (Maity, Kumarasamy, & Razak, 2025). Movement of farm equipment increases the risk of potential electric shocks and damaging solar panels. Safety elements used for these risks implement an effective cable management system that does not disturb equipment (SB Staff, 2025). Dust deposition is another risk. Panels are only cleaned with water; immoderate spraying water can lead to an increase in runoff and soil erosion. Implementing proper panel tilting can mitigate water distribution and reduce the risk of runoff and soil erosion (Zahrawi & Aly, 2024). In addition, dust reduces power production, designing a suitable tilted and angled photovoltaic system will minimize soiling of the systems (Ghosh, 2023). All these safety elements have been evaluated with constraints and performed in the criterium that have been used for selecting the conceptual design of this project.

Conceptual Design

A brief component of agrivoltaic designs is outlined in this section, a detailed layout and decision matrix is attached in appendix B. The agrivoltaics configurations studied are fixed tilt, vertical bifacial, and single axis tracking. Each type has benefits and detriments that impact energy production, crop yield, and profitability. Fixed tilt systems are characterized by having a constant position of their photovoltaic field (Barbon et al., 2025). Vertical bifacial systems are characterized by having the panels at 90°. A single axis tracker rotates its photovoltaic field from

east to west, following the movement of the sun. The orientation of the panel must have its axis of rotation aligned with the north-south axis.

Recommended Design Option

Vertical bifacial is the recommended agrivoltaic array configuration based on its performance across key evaluation criteria. The weighted decision matrix compares three designs: fixed tilt, single axis tracking, and vertical bifacial using seven metrics: crop impact, energy output, cost, environmental impact, land use efficiency, profitability, and scalability. As shown in Table B1, vertical bifacial systems received the highest total score of 55, outperforming fixed tilt 48 and single axis tracking 29. This ranking reflects vertical bifacial's ability to perform well in, environmental impact, land use efficiency, crop impact, and scalability. Vertical bifacial systems use panels mounted at 90 degrees. They typically use an east-west orientation. It has been observed that these systems can maximize land use and provide reliable performance across seasons. They benefit from diffused irradiance and can maintain crop productivity beneath and around the panel rows.

This configuration produces a camel-shaped generation profile, two peaks in the morning and afternoon, that align more closely with electricity demand patterns and can improve market value in systems dominated by conventional PV (Chuinzow et al., 2020). Vertical bifacial systems align closely with demand and the seven metrics in the decision matrix. It is for these reasons that it is the recommended design.

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Appendix A: Regional Focus and District Composition

Table A1. Michigan Districts of Focus and Included Counties.

District 7	Allegan, Barry, Kent, and Ottawa
District 8	Clinton, Eaton, Gratiot, Ingham, Ionia, and Montcalm
District 13	Branch, Berrien, Cass, Calhoun, Kalamazoo, St. Joseph, Van Buren

Figure A1. Map of Michigan Districts.

Appendix B: Design Alternatives and Decision Matrix

Justification of Designs

Within each configuration, there are certain factors to consider that impact on the overall designs. An important factor to consider is whether the fixed tilt and horizontal single axis tracking are monofacial or bifacial. However, bifacial panels lead to a significant increase in energy yield compared to monofacial panels (Guerrero-Lemus et al., 2016). Furthermore, the

cost of bifacial panels should not cost significantly more than monofacial panels given today's state-of-the-art panel modules (Berrian et al., 2019).

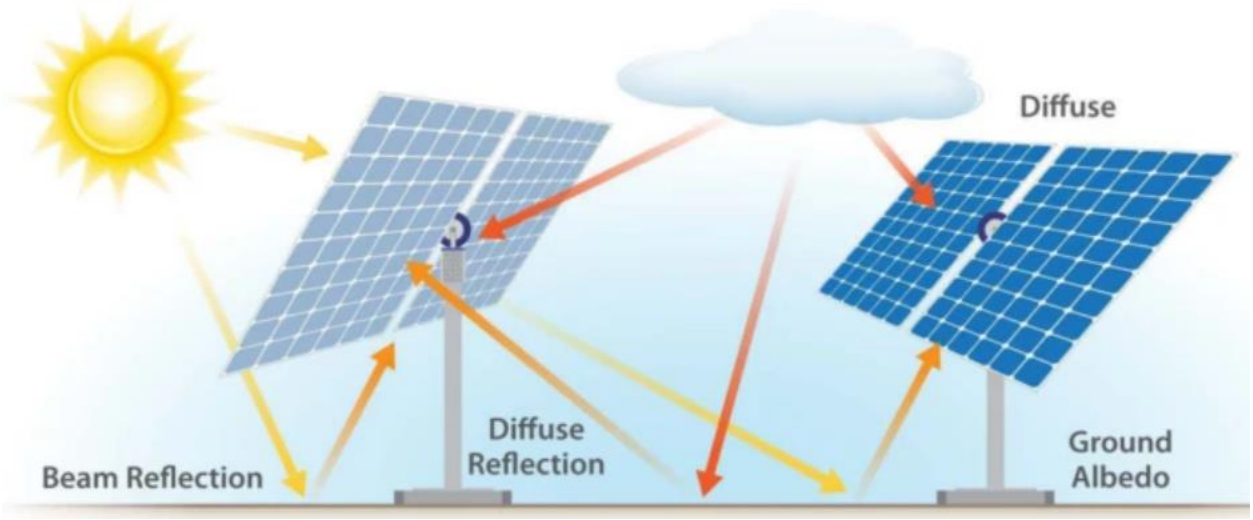

Figure 2: Illustration of bifacial photovoltaic system and the different paths of sunlight take to reach the panels (Badran et al., 2024)

Another factor to consider when comparing different panel configurations is photosynthetically active radiation (PAR) under the solar panels. The shade patterns under photovoltaic panels vary seasonally and spatially depending on the latitude. This means that the shade created by photovoltaic systems will have varying effects on different crops. The availability of PAR, and therefore the difference in panel density and configuration, play crucial roles in the success of crop production un agrivoltaic systems (Prakash et al., 2023).

Within agrivoltaics, elevated systems have begun to see research and implementation in certain regions, practically Europe. Elevated systems see a greater cost of efficiency in electricity generation. Despite their greater efficiency to generate electricity, elevated systems have a significantly higher levelized cost of electricity and saw significantly lower profitability and cost-effectiveness (Zidane et al., 2025).

Fixed Tilt

Fixed tilt systems are cheaper than other systems, which makes them a favorable consideration when implementing photovoltaics in locations that are financially constrained. Fixed- tilt systems can be easily compared to vertical systems as they are both stationary. The key difference between the two is the angle and the orientation of the panels. Fixed tilt systems have been found to work best with a North/South orientation, with panels facing south (Riaz et al., 2021). Fixed tilt systems can be both mono-facial and bifacial. Considering mono-facial systems, they work best for energy production at about 40° (Riaz et al., 2021). However, considering crop growth and albedo for bifacial systems, the best angle for the greatest global ground radiation is 75° as it creates less shading on the ground (Riaz et al., 2021). The tilt angle on fixed systems can be manually adjusted to specific angles for each month to capture the most solar radiation (Prakash et al, 2023). Bifacial systems increase the power density compared to mono-facial systems, especially in conditions with a high albedo (R. Guerrero-Lemus 2016). Mono-facial fixed tilt systems work best closer to the equator while tracking and vertical systems work better at higher latitudes (Riaz et al., 2021).

Figure 3: Example of a fixed tilt agrivoltaic system installation in southern New Jersey

Vertical Bifacial

Vertical bifacial systems are characterized by having the panels at 90°.Vertical are typically set up with an East/West orientation. Vertical bifacial has seen significant research and development in European countries. This is because it has been found that at latitudes of 50° and above, vertical bifacial photovoltaic systems generate more electricity than conventional photovoltaic systems (Chudinzow et al., 2020). Vertical bifacial photovoltaic systems have been found to have a great ability to harness solar energy regardless of seasonal variances and can help to maximize land use. Bifacial photovoltaic panels see a higher yield of energy because of diffused irradiance, which is solar radiation reflecting on particles in the atmosphere. Vertical bifacial systems benefit greatly in energy production because of the high amounts of diffused irradiance in the photovoltaic field (Badran et al., 2024)

Figure 4: Example of vertical bifacial photovoltaic farm developed by Next2Sun in Germany

Single Axis Tracking

For a single axis tracking system, the rotational movement of the panel will have a limited tilt angle range, usually ±60° (Barbon et al., 2025). The tracking system would work based on different algorithms. A common algorithm is full tracking which includes standard tracking and back-tracking at low sun angles to minimize shading. Smart tracking combines tracking and anti-

tracking periods. This combination allows tracking to focus on energy yields while anti-tracking can focus on crop yields. This trade-off is achieved based on daily photosynthetic and energy production. Full tracking works best during hot and dry summers to protect crops, and smart tracking works well during rainy years to mitigate crop losses (Willockx et al., 2023). Vertical systems and smart tracking systems have nearly equal amounts of photosynthetically active radiation; however, there are greater agricultural yields with smart tracking algorithms (Willockx et al., 2023).

Figure 5: Example of a single axis tracking system in Grembergen, Belgium. This example shows the land lost to buffer zones for the photovoltaic system.

Decision Matrix

Photovoltaic systems offer farmers a promising way to generate additional income from solar energy. However, selecting the right array configuration is critical. Fixed tilt systems are the most affordable and simplest to install, making them ideal for budget-conscious farms. They perform well with south-facing panels and bifacial designs that reduce shading and improve ground-level light. Vertical bifacial systems, with panels mounted at 90° in an east-west orientation, excel in land-use efficiency and seasonal reliability, especially at higher latitudes.

They minimize shading and support consistent crop growth. Single axis tracking systems rotate to follow the sun, offering the highest energy yield, particularly when using smart tracking algorithms that balance energy production with crop health. However, they come with higher and more complex maintenance.

Determining that the agrivoltaic designs generate enough profit to justify implementation is one of the main goals of this project. This can be done by calculating the Net Present Value (NPV) and the levelized cost of electricity (LCOE). The parameter that will most affect the NPV of a photovoltaic system is the average profit of crops (Zidane et al., 2025). LCOE is mainly influenced by the cost of the photovoltaic system, maintenance cost, and the real weighted cost of capital (Zidane et al., 2025). LCOE can be calculated using the following equation:

$$LCOE = \frac{I_0 + \sum_{t=1}^{n} \frac{A_t}{(1+i)^t}}{\sum_{t=1}^{n} \frac{M_{t,el}}{(1+i)^t}} \qquad (2)$$

Where I_0 is the initial investment at year zero, A_t is the annual cost in year t, $M_{t,el}$ is the electricity generated in year t, n is lifetime of the photovoltaic system, and i is the interest rate.

Performance Criteria

The performance criteria for the agrivoltaics system were determined from the constraints. Find below a list of criteria used to determine the effectiveness of the potential designs:

- Cost: upfront and long-term expenses: installation, material, labor
- Profitability: financial return generated
- Scalability: system adaptability across acreage
- Environmental Impact: runoff, water distribution, erosion, wind flow, and heavy snow
- Land Use Efficiency: agricultural and energy efficiency per acre
- Crop Impact: panel layout affecting crop, shading, and temperature
- Energy Output: energy production

Table B1. Decision matrix for 3 alternatives and 7 parameters.

	Crop Impact	Energy Output	Cost	Environmental Impact	Land Use Efficiency	Profitability	Scalability	Weighted Totals
Importance (weighting)	18	18	15	15	14	10	10	100
Fixed Tilt	5	3	3	3	3	3	3	48
Single Axis Tracking	1	5	5	1	1	5	1	29
Vertical Bifacial	3	1	1	5	5	3	5	55

Energy, maintenance, cost, land use efficiency + crop, environmental impact (heavy snow, lowest footprint means lowest erosion) (soil erosion due to runoff) lowest footprint better, Comparing these designs, it confirms that vertical bifacial systems offer the best overall performance, scoring highest in crop impact, land-use efficiency, and equipment accessibility. Fixed tilt systems ranked second, with strong scores in cost, maintenance, and energy output. While single axis tracking systems lead in energy yield and profitability but scored lowest overall due to reduced accessibility and higher operational demands. These insights will guide the optimization model in recommending array configurations that align with farm profiles, operational constraints, and financial goals ensuring farmers can maximize income.

Appendix C: Project Management Summary and Work Breakdown Structure

This project will be divided into four phases. The work will move from farmer engagement and literature review to the development of an optimization model that will allocate land between crop production and photovoltaic (PV) energy generation on Michigan farms. The optimization model will be the primary output. A written report describing the model, its constraints, design alternatives, and evaluation will serve as complementary output. Work will involve Michigan State University Extension (MSUE), the Southwest Michigan Research and Extension Center (SMREC), and growers in Districts 7, 8, and 13 and will span two semesters: BE 485 (Fall 2025) and BE 487 (Spring 2026).

Phase 1 – Background, Standards, and Stakeholder Engagement

(Fall 2025, Weeks 1–7)

Phase 1 will align with the BE 485 startup period and early writing milestones. The team will build an understanding of the project, meet with the faculty advisor and client, and complete early literature review and project summary assignments that lead to the Project Overview. The team will review literature on agrivoltaics, soil health, crop rotations, economics, social acceptance, ecological impacts, and siting and zoning rules in Michigan. The team will map key national and state standards for quality management, worker safety, electrical design, food safety, and land use, and summarize them for Appendix A. The team will begin coordination with MSUE, SMREC, and growers in Districts 7, 8, and 13.

Phase 2 – Data Collection, Case Studies, and Co-Design with Growers

(Fall 2025, Weeks 8–15)

Phase 2 will align with Industry Advisory Board input, and the design alternatives assignment. The team will refine the Project Overview and design alternatives based on feedback and will sharpen the problem statement, objectives, constraints, and performance criteria for the model. The team will hold structured conversations with farmers in Districts 7, 8, and 13 to describe cropping systems, field layouts, equipment access, management practices, and crop calendars to identify major opportunities and barriers for agrivoltaic layouts. The team will compile Web Soil Survey data and any site-specific soil tests to describe representative soil conditions. By the end of Week 15, the team will have farm profiles, priority crop rotations, candidate layouts, and a consolidated set of constraints for BE 485 Faculty Jury review as the basis for optimization in BE 487.

Phase 3 – Optimization Model Development

(Spring 2026, Weeks 1–10)

Phase 3 will take place in BE 487 during the first ten weeks of the semester. The team will choose the final design approach, prepare an initial design report for review, revise it, and implement the optimization model. The team will convert siting, soil, equipment, environmental, financial, and time constraints from Phases 1 and 2 into model parameters and decision variables. The model will include constraints for array layout, machinery access, interconnection, site conditions, and financial limits and will determine how to allocate land between crop production and PV generation to maximize profit while maintaining soil health and practical operation. Model outputs will summarize energy generation, crop yield, economic performance, land-use efficiency, access and maintenance needs, and environmental indicators. The team will calibrate and validate the model using Michigan solar installer data, agrivoltaic case studies, and grower

information gathered in BE 485 and will evaluate candidate designs with a decision matrix to select a recommended configuration and refine model scenarios.

Phase 4 – Synthesis, Recommendations, and Final Deliverables

(Spring 2026, Weeks 11–15 and Final Exam Week)

Phase 4 will occupy the final third of BE 487. The team will complete the final design report, share it with the Industry Board, present it to the faculty jury, and prepare the final poster, BE Showcase presentation, and Design Day poster. The team will synthesize model results, decision matrix scores, standards, safety requirements, and farmer feedback into design recommendations for agrivoltaic layouts on Michigan specialty-crop and cash-crop farms. The recommendations will specify preferred array configurations, spacing strategies, crop rotations, and operating guidelines and will aim to protect soil health, maintain visible agriculture, support energy production, and improve farm income while meeting Michigan siting rules. The team will finalize the optimization model as a decision-support tool for farmers, developers, and MSUE and will finalize the written report, the standards and regulations appendix, and supporting figures, tables, and equations. The BE Showcase and Design Day materials will communicate the problem, model structure, key findings, and recommended next steps to faculty and industry evaluators. A work breakdown structure (WBS) in Appendix E will list major tasks, subtasks, timing, and responsibilities for BE 485 and BE 487 and will guide schedule, budget, and alignment with course expectations and client needs.

Table C1. Work breakdown of effort as a group.

Task Name	Duration	Start	Finish	Resource Names
1 Project Startup				**Project Startup**

1.1 Initial team meetings and background research				**Initial Meetings and Research**
1.1.1 Team Meeting	1 hr	Mon 9/22/25	Mon 9/22/25	Introductory meeting
1.1.2 Team Meeting	0.5 hrs	Mon 9/29/25	Mon 9/29/25	Introductory meeting
1.1.3 Team Meeting	4 hrs	Sat 10/4/25	Sat 10/4/25	Proposed project draft
1.2 First client meeting				
1.2.1 First client meeting	1.75 hrs	Fri 10/10/25	Fri 10/10/25	Client expectations
1.2.1 Team Meeting	1.75 hrs	Sun 10/12/25	Sun 10/12/25	Client meeting debriefs
2 Project Selection				
2.1 Informative Meetings				**Informative Meetings**
2.1.1 Instructor Meeting	0.3 hrs	Wed 10/1/25	Wed 10/1/25	Review of expectations
2.1.2 Team Meeting	0.5 hrs	Wed 10/1/25	Wed 10/1/25	Debrief meeting
2.1.3 Instructor Meeting	0.17 hrs	Wed 10/8/25	Wed 10/8/25	Proposed project feedback
2.1.4 Team Meeting	0.6 hrs	Wed 10/8/25	Wed 10/8/25	Debrief meeting
2.2 Proposed Project Summary				**Propose Project Summary**
2.2.1 Team Meeting	1 hr	Mon 10/6/25	Mon 10/6/25	Report #3 edits
2.2.2 Team Meeting	1 hr	Mon 10/13/25	Mon 10/13/25	Report #3 edits
2.2.3 Instructor Meeting	0.3 hrs	Wed 10/15/25	Wed 10/15/25	Report #3 update
2.2.4 Team Meeting	1 hr	Wed 10/15/25	Wed 10/15/25	Feedback debrief meeting
2.3 External Feedback Meetings				**External Feedback Meetings**
2.3.1 Industry Advisory Board Meeting	0.5 hrs	Thu 10/16/25	Thu 10/16/25	External Feedback
2.3.2 Team Meeting	0.5 hrs	Tue 10/14/25	Tue 10/14/25	Project preparation
2.4 Assignment #4 Project Overview				**Assignment #4 Project Overview**
2.4.1 Team Meeting	1.16 hrs	Mon 10/27/25	Mon 10/27/25	Report #4 draft
2.4.2 Team Meeting	1.5 hrs	Wed 10/29/25	Wed 10/29/25	Report #4 edits
2.4.3 Team Meeting	0.5 hrs	Thu 10/30/25	Thu 10/30/25	Report #4 edits
2.4.4 Instructor Meeting	0.3 hrs	Wed 11/5/25	Wed 11/5/25	Report #4 feedback
2.4.5 Teem Meetings	0.7 hrs	Wed 11/5/25	Wed 11/5/25	Feedback debrief meeting
2.5 Project Realignment				**Project realignment**
2.5.1 Call with Client	0.5 hrs	Tue 11/4/25	Tue 11/4/25	Client report feedback
2.5.2 Team Meeting	0.3 hrs	Tue 11/4/25	Tue 11/4/25	Client realignment

2.5.3 Secondary Faculty Meeting	0.7 hrs	Wed 11/5/25	Wed 11/5/25	Advisor realignment feedback
2.5.4 Primary Faculty Meeting	1 hr	Thu 11/6/25	Thu 11/6/25	Advisor realignment feedback
2.5.5 Team Meeting	0.3 hrs	Fri 11/7/25	Fri 11/7/25	Feedback debrief meeting
2.5.6 Team Meeting	3 hrs	Sat 11/8/25	Sat 11/8/25	Multi-work session
2.5.7 Client Overview	1 hr	Wed 11/12/25	Wed 11/12/25	Client realignment meeting
3 Introductory Project Presentation				**Introductory Project Presentation**
3.1 Team Meeting	0.7 hrs	Fri 10/17/25	Fri 10/17/25	Presentation edits
3.2 Team Meeting	0.8 hrs	Wed 10/22/25	Wed 10/22/25	Presentation edits
3.3 Team Meeting	2 hrs	Fri 10/24/25	Fri 10/24/25	Presentation edits
3.4 Team Meeting	0.5 hrs	Sun 10/26/25	Sun 10/26/25	Final presentation edits
3.5 Presentation Feedback Instructor Meeting	0.16 hrs	Mon 11/3/25	Mon 11/3/25	Presentation feedback
3.6 Team Meeting	1.5 hrs	Mon 11/3/25	Mon 11/3/25	Feedback debrief meeting
4 Project Design				**Project Design**
4.1 Assignment #6 Design Alternatives and Decision Matrix				**Assignment #6**
4.1.1 Team Meeting	1.75 hrs	Mon 11/10/25	Mon 11/10/25	Report #6 edits
4.1.2 Team Meeting	1.3 hrs	Wed 11/12/25	Wed 11/12/25	Report #6 edits
4.1.3 Instructor Meeting	0.3 hrs	Wed 11/12/25	Wed 11/12/25	Instructor updates
4.1.4 Team Meeting	3 hrs	Fri 11/14/25	Fri 11/14/25	Report #6 edits
4.1.5 Team Meeting	6 hrs	Sun 11/16/25	Sun 11/16/25	Multi-work session
4.1.6 Faculty Meeting	1.5 hrs	Mon 11/17/25	Mon 11/17/25	Advisory updates
4.1.7 Team Meeting	1 hr	Mon 11/17/25	Mon 11/17/25	Advisory updates
4.1.8 Team Meeting	2 hrs	Tue 11/18/25	Tue 11/18/25	Report #6 edits
4.1.9 Instructor Meeting	0.3 hrs	Wed 11/19/25	Wed 11/19/25	Instructor updates
4.1.10 Team Meeting	2 hrs	Wed 11/19/25	Wed 11/19/25	Report #6 edits
4.1.11 Individual Work	15 hrs	Tue 11/18/25	Wed 11/19/25	Each individual works on report #6 final edits

Appendix D: Project Budget Projections

Phase 1 – Budget $250

Expense – Hard Hat with Cap Brim (4-Point Ratchet Suspension, orange, 4) – $79.40
Expense – High-Visibility ANSI 2 Mesh Vests, Yellow, Size L (3) – $48.42
Expense – High-Visibility ANSI 2 Mesh Vest, Yellow, Size XL (1) – $16.14
Expense – Shipping for PPE order – $20.35

Remaining – $85.69

Phase 2 – Budget $250

Expense – Soil testing, 5 samples @ $26 each – $130
Expense – Travel mileage for field visits, 100 miles @ $0.67/mile – $67
Expense – Farmer meeting refreshments (snacks and drinks) – $30

Remaining – $23

Phase 3 – Budget $250

Expense – Intermediate design review poster printing (1 research poster) – $25
Expense – Printing and office supplies for draft reports and figures – $50

Remaining – $175

Phase 4 – Budget $250

Expense – Final Design Day research poster – $25
Expense – BE Showcase / jury poster – $25
Expense – Color printing and binding of final reports – $50
Expense – Display materials for presentations (foam board, tape, mounting supplies) – $30

Remaining – $120

Total Remaining – $403.69